

The 3D porous metal–organic frameworks based on bis(pyrazinyl)–triazole: structures, photoluminescence and gas adsorption properties†

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Two novel 3D porous metal–organic frameworks, $\{[\text{Zn}_2(\text{bpt})(\text{btc})] \cdot 4\text{H}_2\text{O}\}_n$ (**1**) and $\{[\text{Cd}_2(\text{bpt})(\text{btc})(\text{H}_2\text{O})] \cdot (\text{DMA})_2\}_n$ (**2**) (DMA = *N,N*-dimethylacetamide), were prepared based on 3,5-bis(pyrazinyl)-1,2,4-triazole (Hbpt) with the auxiliary ligand 1,3,5-benzenetricarboxylic acid (H_3btc). Complex **1** could be described as a (3,6)-connected binodal 3D network with rutile topology and Schläfli symbol of $(4.6^2)_2(4^2.6^{10}.8^3)$, where its coordination framework exhibits a high thermal stability up to 420 °C as well as excellent boiling-water resistance. Complex **2** shows a (3,6,8)-connected trinodal 3D framework with a Schläfli symbol of $(4.6^2)_2(4^{14}.6^{12}.8^2)(4^4.6^{10}.8)$, which has not been reported previously. Interestingly, the gas-adsorption investigation discloses that complex **1** possesses a high enthalpy of H_2 adsorption (9.6 kJ mol^{−1}). Intensive solid-state fluorescence spectra of complexes **1** and **2** have also been determined.

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Introduction

Porous metal–organic frameworks (MOFs), emerging as a new class of functional material, have been widely studied not only owing to their topologically interesting architectures but also potential applications in gas storage/separation, optics, catalysis, sensing, and so on.¹ Numerous studies have been reported in the past few years relating to hydrogen storage since hydrogen is regarded as a clean energy that could replace fossil fuels.² It is more convincing that some new materials of MOFs can store a large amount of hydrogen. For example, the hydrogen storage capacities of SNU-77 and MOF-210 are 8.1 wt% (total 11.0 wt%) at 77 K and 90 bar,³ and 8.6 wt% (total 16.7 wt%) at 77 K and 70 bar,⁴ respectively. However, the practical applications of the MOFs as hydrogen storage are very much diminished at ambient temperature. Thus, the isosteric heat (Q_{st}) of H_2 adsorption should be substantially increased to facilitate H_2 adsorption at ambient temperature and pressures. One somewhat successful strategy has been constructing the frameworks with narrow pores to increase the

affinity for H_2 adsorption by using relatively small ligands in the hybrid synthesis.⁵

On the other hand, coordination polymers constructed by N-donor ligands have been widely reported in recent years.⁶ Specifically, 1,2,4-triazole and its derivatives, which combine the coordination geometry of both pyrazoles and imidazoles, are very interesting as bridging ligands.⁷ The 1,2,4-triazole based ligands with pyridyl groups have been employed in the construction of novel architectures.⁸ However, the coordination polymers based on 1H-3,5-bis(pyrazinyl)-1,2,4-triazole (Hbpt), where the 1H-1,2,4-triazole moieties are introduced between the two pyrazine groups, are have not been reported. As a versatile tecton, Hbpt contains seven potential N-donors, which can be used as a uni-, bidentate, or tridentate connector to construct diverse metallosupramolecular systems *via* different coordination modes.

Furthermore, it is a rigid angular multifunctional ligand, which is expected to construct many high-dimensional and highly connected coordination frameworks. In order to take advantages of different types of ligands, we chose a polycarboxylic ligand, H_3btc (1,3,5-benzenetricarboxylic acid), as an O-donor auxiliary ligand to synthesize complexes with diverse structures and topologies as well as specific properties.

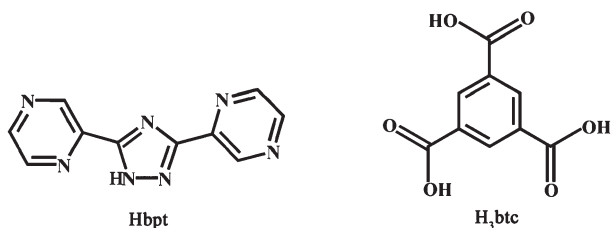
Considering the aforementioned points, we successfully obtained two non-interpenetrated 3D microporous frameworks, $\{[\text{Zn}_2(\text{bpt})(\text{btc})] \cdot 4\text{H}_2\text{O}\}_n$ (**1**) and $\{[\text{Cd}_2(\text{bpt})(\text{btc})(\text{H}_2\text{O})] \cdot (\text{DMA})_2\}_n$ (**2**), derived from Hbpt and H_3btc (Scheme 1) with Zn^{II} / Cd^{II} . The crystal structures, topological analyses, gas adsorption and luminescent properties for both **1** and **2** have been well

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† Electronic supplementary information (ESI) available: The selected bond lengths (Å) and bond angles (°) for complexes **1** and **2**, XRD patterns of complex **2**, and some additional figures. CCDC 926786 and 926787 for complexes **1** and **2**. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c3ce40574e



Scheme 1 The ligands 3,5-bis(pyrazinyl)-1,2,4-triazole (Hbpt) and 1,3,5-benzenetricarboxylic acid (H₃btc).

investigated. More interestingly, complex **1** can be considered as a success of the above approaches to construct a microporous material with narrow pores, which exhibits a remarkable isosteric heat of adsorption for H₂.

Experimental

Materials and methods

All reactants were reagent grade and used as purchased without further purification. Elemental analyses for C, H, and N were carried out on a German Elementary Vario EL III instrument. The FT-IR spectra were performed on a Nicolet Magna 750 FT-IR spectrometer using KBr pellets in the range of 4000–400 cm^{−1}. Thermogravimetric analyses were recorded on a NETZSCH STA 449C unit at a heating rate of 10 °C min^{−1} under a nitrogen atmosphere. The power X-ray diffraction (XRD) patterns were collected by a Rigaku DMAX2500 X-ray diffractometer using Cu Kα radiation (λ = 0.154 nm). Fluorescent analysis of the complexes was performed on a Horiba Jobin–Yvon Fluorolog-3 spectrophotometer analyzer.

Measurement of gas adsorption

N₂ and ambient-pressure H₂ isotherms at 77 K for complexes **1** and **2** were maintained by a liquid nitrogen bath, with pressure ranging from 0 to 1 bar on ASAP (Accelerated Surface Area and Porosimetry) 2020 System, while CO₂ and CH₄ adsorption measurements were done at 273 K/300 K in the same instrument at the same pressure range. The freshly prepared samples of **1** and **2** were soaked in methanol for 3 days to remove free solvates present in the frameworks and then outgassed overnight under vacuum to generate the dehydrated form **1a** and **2a**.

Synthesis of {[Zn₂(bpt)(btc)]·4H₂O}_n (**1**)

A mixture of Zn(NO₃)₂·6H₂O (0.6 mmol, 179 mg), Hbpt (0.1 mmol, 22 mg), H₃btc (0.1 mmol, 21 mg), and 10 mL of H₂O–EtOH (1/1, v/v) was sealed in a 23 mL Teflon-lined stainless steel autoclave and heated at 120 °C for 3 days. After the mixture was cooled to room temperature, pale yellow block crystals of **1** were obtained with the yield of 89% based on Hbpt. Anal. calcd for C₁₉H₁₇N₇O₁₀Zn₂ (fw 634.14): C, 35.98%; H, 2.69%; N, 15.46%. Found: C, 35.34%; H, 2.98%; N, 15.35%. Selected IR data (KBr pellet, cm^{−1}): 3465(s), 3101(s), 2967(w), 1619(s), 1560(s), 1437(s), 1375(s), 1153(w), 1066(w), 936(w), 861(w), 766(m).

Synthesis of {[Cd₂(bpt)(btc)(H₂O)]·(DMA)₂}_n (**2**)

Complex **2** was synthesized from Cd(NO₃)₂·4H₂O (0.6 mmol, 185 mg), Hbpt (0.1 mmol, 22 mg), H₃btc (0.1 mmol, 21 mg), and 10 mL of H₂O–DMA (1/1, v/v, DMA = *N,N*-dimethylacetamide). The mixture was sealed in a 23 mL Teflon-lined stainless steel autoclave and heated at 120 °C for 3 days. After it was cooled down to room temperature, light yellow block crystals of **2** were separated by filtration with the yield of 68% based on Hbpt. Anal. calcd for C₂₇H₂₉N₉O₉Cd₂ (fw 848.39): C, 38.22%; H, 3.45%; N, 14.86%. Found: C, 37.98%; H, 3.26%; N, 14.98%. Selected IR data (KBr pellet, cm^{−1}): 3060(w), 2933(w), 1622(s), 1554(s), 1433(s), 1372(s), 1148(w), 1024(w), 866(w), 760(m), 726(m).

X-Ray crystallography

All the structural data of complexes **1** and **2** were collected on a Rigaku Mercury CCD diffractometer equipped with a graphite-monochromated Mo Kα radiation (λ = 0.71073 Å) at room temperature. All of the structures were resolved by the direct method and refined by full-matrix least-squares fitting on *F*² by SHELX-97.⁹ All non-hydrogen atoms were refined with anisotropic thermal parameters. The positions of hydrogen atoms on the organic ligands were generated geometrically and refined using a riding model. In the case of **1**, the hydrogen atoms of O4 were localized on a difference Fourier map and further refined by applying a restraint on the O–H bond lengths. In the case of **2**, hydrogen atoms of O8 could not be determined. In the structure of complex **1**, two water molecules were located in the difference electron density map while two additional highly disordered water molecules were treated with the program PLATON/SQUEEZE. The final chemical formula of **1** was calculated from SQUEEZE results combined with the TGA and elemental analyses data. In addition, the relatively large ADP values for the DMA molecules in **2** might be due to no obvious strong interactions between the free DMA molecules and the host framework. Crystallographic data and structure refinement parameters for complexes **1** and **2** are summarized in Table 1. Selected bond lengths and bond angles are listed in the ESI† More details on the crystallographic studies as well as atomic displacement parameters are given in the ESI† as CIF files. CCDC 926786 and 926787 contain the supplementary crystallographic data for this paper.

Results and discussion

Crystal structure of {[Zn₂(bpt)(btc)]·4H₂O}_n (**1**)

A single-crystal X-ray diffraction analysis reveals that complex **1** crystallizes in the tetragonal space group *P*₄₃₂₁₂ and exhibits a dinuclear Zn₂-based non-interpenetrated 3D framework. The asymmetric unit is composed of one Zn(II) ion, half of a deprotonated bpt[−] ligand, half of a btc^{3−} anion, and two free water molecules in a crystal lattice. As shown in Fig. 1, the Zn center is five-coordinated with two nitrogen atoms (N4, N1B) from two bpt[−] ligands and three carboxylate oxygen atoms (O1, O2C, O3C) from different btc^{3−} ligands, resulting in a distorted square pyramid geometry, in which N4, N1B, O2C

Table 1 Crystal data and structure refinements for complexes **1** and **2**

Complexes	1	2
Empirical formula	C ₁₉ H ₁₇ N ₇ O ₁₀ Zn ₂	C ₂₇ H ₂₉ N ₉ O ₉ Cd ₂
Formula weight	598.10	846.38
Crystal system	Tetragonal	Monoclinic
Space group	<i>P</i> 4 ₃ 2 ₁ 2	<i>C</i> 2/ <i>c</i>
<i>a</i> [Å]	10.8140(13)	20.730(8)
<i>b</i> [Å]	10.8140(13)	14.751(5)
<i>c</i> [Å]	25.410(4)	20.709(8)
α [°]	90.00	90.00
β [°]	90.00	97.174(4)
γ [°]	90.00	90.00
<i>V</i> [Å ³]	2971.5(7)	6283(4)
<i>Z</i>	4	8
<i>D_c</i> [g cm ⁻³]	1.337	1.790
μ [mm ⁻¹]	1.66	1.421
<i>F</i> (000)	1200	3360
θ range [°]	3.1–27.5	2.5–27.5
Collected reflections	22 870	18 081
Unique reflections	3397	5836
GOF on <i>F</i> ²	1.096	1.081
Parameters	160	432
<i>R</i> ₁ [<i>I</i> > 2 σ (<i>I</i>)] ^a	0.0319	0.0626
<i>wR</i> ₂ [<i>I</i> > 2 σ (<i>I</i>)] ^b	0.0879	0.2053
Flack parameter	0.276(18)	

$$^a R_1 = \sum ||F_o| - |F_c|| / \sum |F_o|; \quad ^b wR_2 = [\sum w(F_o^2 - F_c^2)^2 / \sum w(F_o^2)^2]^{1/2}.$$

and O3C atoms locate at the bottom plane, and the axial position is occupied by the O1 atom. The distances of Zn–N bonds range from 2.019 Å to 2.119 Å, and the Zn–O bond lengths are in the range of 1.975–2.403 Å, which are comparable to the related MOFs.¹⁰ Interestingly, one deprotonated bpt[−] ligand bridges two Zn²⁺ ions to form a dinuclear unit and the distance between the dinuclear Zn²⁺ ions is 4.031 Å. Moreover, the dinuclear units are interconnected by btc^{3−} ligands *via* Zn–O bonds to form a 3D metal–organic framework with 1D tetragonal channels (Fig. 2). The sizes of

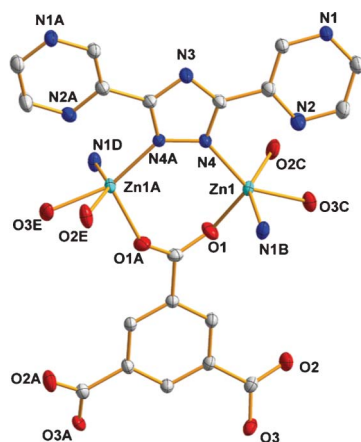


Fig. 1 The coordination environment of the Zn(II) ion in **1** with the thermal ellipsoids at 30% probability level. Symmetry codes (A) *x*, *y*, *−z*; (B) $-1/2 + x$, $3/2 - y$, $1/4 - z$; (C) $1/2 + x$, $1/2 - y$, $1/4 - z$; (D) $-1/2 + x$, $3/2 - y$, $-1/4 + z$; (E) $1/2 + x$, $1/2 - y$, $-1/4 + z$. (The H atoms and water molecules are omitted for clarity.)

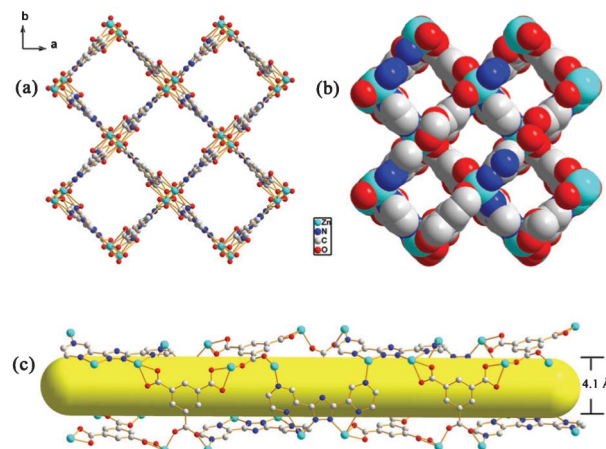


Fig. 2 (a) Perspective view of the 3D framework of complex **1** along the [001] direction; the guest water molecules are omitted for clarity; (b) space-filling model of complex **1** along the *c* axis showing the large 1D tetragonal channels; the guest water molecules are omitted for clarity; (c) 1D tetragonal channel with a pore diameter of 4.1 Å.

the channels are about 4.1×4.1 Å². The solvent accessible volume is estimated by the PLATON program to be about 43.3% of the total crystal volume. The void is occupied by four free water molecules per formula unit in **1**, which is confirmed by TGA and elemental analyses for crystal samples.

To best understand the structure of complex **1**, the topology analysis is employed to describe the architecture in detail. Each dinuclear unit is connected by three bpt[−] ligands and three btc^{3−} ligands, thus, it can be considered as a six-connected node, while the bpt[−] and btc^{3−} ligands involving the metal cations chelated by them are considered three-connected nodes. According to the simplification principle, the resulting structure is a binodal (3,6)-connected three-dimensional network with rutile topology and the Schläfli symbol of $(4.6^2)_2(4^2.6^{10}.8^3)$ (Fig. 3).¹¹

Crystal structure of $\{[\text{Cd}_2(\text{bpt})(\text{btc})(\text{H}_2\text{O})](\text{DMA})_2\}_n$ (**2**)

Complex **2** crystallizes in the monoclinic *C*2/*c* space group. In the asymmetrical unit, there are two crystallographically independent Cd(II) ions, one bpt[−] ligand, one btc^{3−} ligand, one coordinated water molecule, and two free DMA molecules.

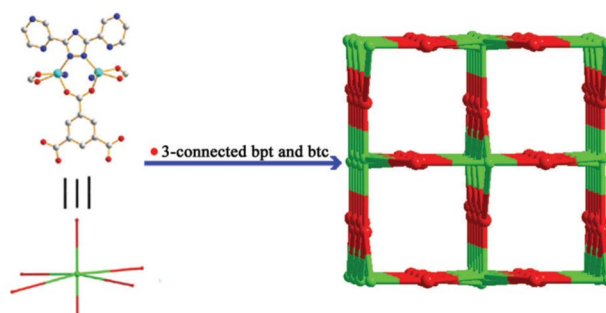


Fig. 3 Rutile topology net of **1**: green, Zn₂ dinuclear unit; red, ligands.

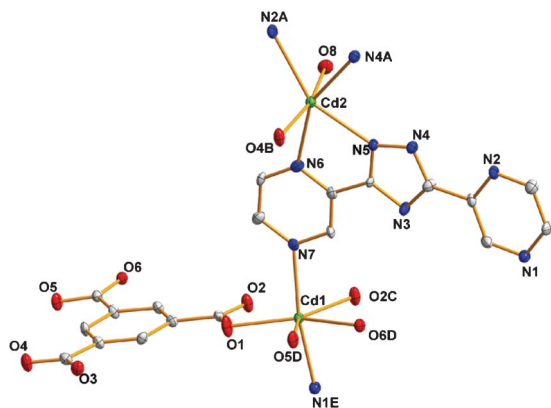


Fig. 4 The coordination environment of the Cd(II) ions in **2** with the thermal ellipsoids at 30% probability level. Symmetry codes (A) $1 - x, -y, -z$; (B) $1/2 - x, -1/2 + y, 1/2 - z$; (C) $1 - x, y, 1/2 - z$; (D) $x, 1 - y, -1/2 + z$; (E) $1 - x, 1 - y, -z$. (The H atoms and DMA molecules are omitted for clarity.)

Each of the two metal cations, Cd1 and Cd2, is six-coordinated in a slightly distorted octahedral environment. As shown in Fig. 4, for Cd1, the equatorial plane contains four oxygen atoms (O1, O5D, O2C, O6D) from three different btc^{3-} ligands, and the two axial positions are occupied by nitrogen atoms (N1E, N7) of two bpt^- ligands. For Cd2, the metal center is chelated by two bpt^- ligands, and the remaining coordination sites are occupied by two oxygen atoms from one btc^{3-} ligand (O4B) and one coordinated water molecule (O8), respectively. The Cd–N bond lengths vary from 2.341 Å to 2.524 Å and the Cd–O distances range from 2.231 Å to 2.443 Å, which are in good agreement with literature values in Cd^{II} complexes.¹² The nonbonding metal–metal distance between Cd1 and Cd2 is 7.554 Å.

If the connections through the btc^{3-} ligands are ignored initially, a $\text{Cd}_2(\text{bpt})_2$ SBU formed by two Cd2 atoms and two bpt^- ligands connects another two adjacent SBUs by two Cd1 atoms to form an infinite hinged chain structure (Fig. 5a). Then, the neighboring chains are further connected by the btc^{3-} ligands to generate a 3D network (Fig. 5b). The noninterpenetrated 3D framework contains 1D tetragonal channels along the c axis (Fig. 5c), having an effective aperture size of $3.8 \times 4.6 \text{ Å}^2$. The guest DMA molecules are encapsulated in the channels. About 43.6% solvent accessible volume is estimated by using PLATON software.

To understand the complicated crystal structure, analysis of network topology is used. In complex **2**, the $\text{Cd}_2(\text{bpt})_2$ motif can be simplified to a 6-connected node, while btc^{3-} anions and the binuclear Cd1 atoms can be regarded as 3- and 8-connected nodes, respectively. Thus the simplified overall structure of **2** is a (3,6,8)-connected three-nodal net with stoichiometry $(3\text{-c})_2(6\text{-c})(8\text{-c})$ (Fig. 6). The Schläfli symbol for the net is $(4.6^2)_2(4^{14}.6^{12}.8^2)(4^4.6^{10}.8)$ calculated by the TOPOS program. Up to now, this new topology has not been reported in the literature.

Thermal properties and X-ray powder diffraction

To investigate the thermal stability of the two complexes, thermogravimetric analysis (TGA) experiments were carried

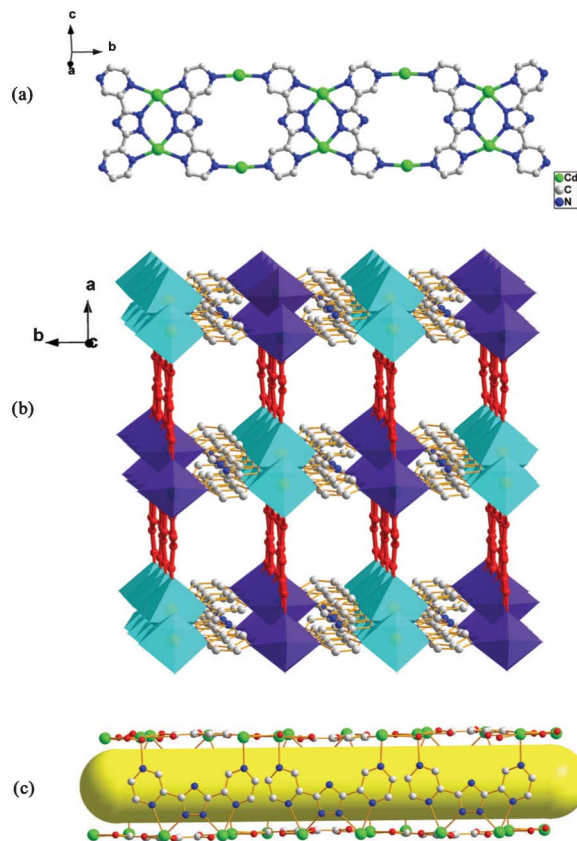


Fig. 5 (a) A drawing showing the hinged chain formed by Cd(II) ions and bpt^- ligands in **2**; (b) a drawing showing the 3D framework (the red spheres represent btc^{3-} ligands, the violet and turquoise polyhedrons represent Cd1 and Cd2 centers, respectively); (c) a drawing showing the 1D tetragonal channel in **2**.

out in the temperature range of 30–1000 °C under a flow of nitrogen with a heating rate of 10 °C min^{-1} (Fig. 7). For complex **1**, the weight loss of 11.03% in the temperature range of 50–420 °C (calc. 11.36%) corresponds to four lattice water molecules, and the decomposition of the residue is observed at about 425 °C, exhibiting a comparatively high thermal stability. The TGA of the methanol-exchanged sample (**1-ex**), prepared by soaking complex **1** in methanol, shows that the guest water molecules can be replaced by methanol (Fig. S1a,

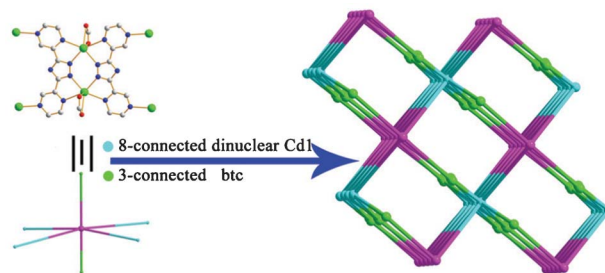


Fig. 6 The (3,6,8)-connected topological net of **2**. The green, turquoise and pink spheres represent btc^{3-} , dinuclear Cd1 atoms, $\text{Cd}_2(\text{bpt})_2$, respectively.

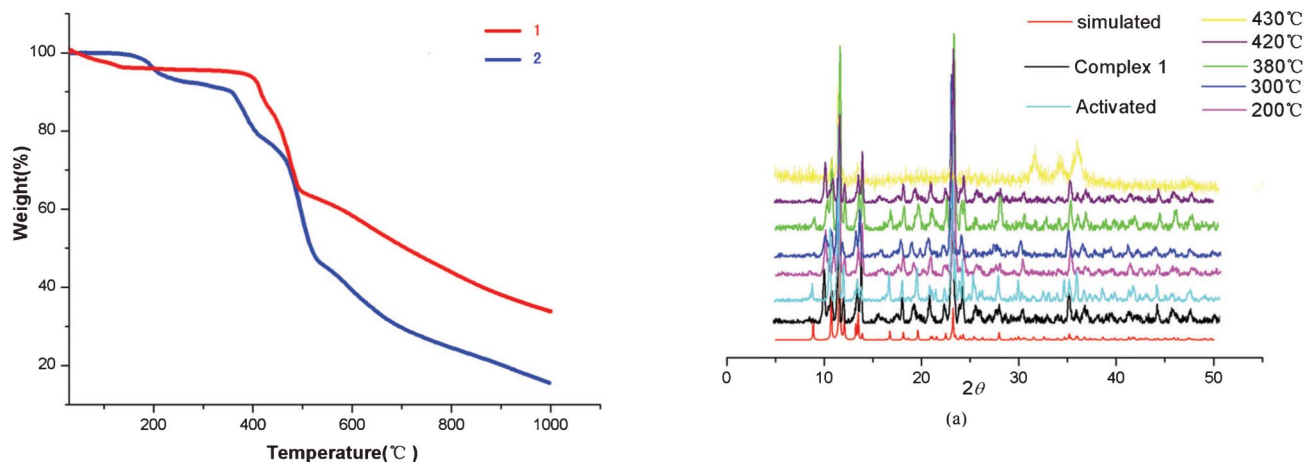


Fig. 7 Thermal gravimetric curves of complexes **1** and **2**.

ESI†). The TGA curve of **2** shows a weight loss of 2.34% below 185 °C, which is ascribed to the release of one coordinated water molecule (calc. 2.12%). From 185 to 400 °C, it loses two free DMA molecules and begins to decompose (calc: 17.61%, found: 17.38%). The TGA of the methanol-exchanged sample (**2-ex**) shows that the guest molecules and the coordinated solvent molecules can be exchanged by methanol (Fig. S1b, ESI†). Subsequently, the samples **1-ex** and **2-ex** are degassed under vacuum, and then the desolvated solids **1a** and **2a** are obtained.

The powder XRD patterns of complexes **1** and **2** in the region of $2\theta = 5\text{--}55^\circ$ are shown in Fig. 8 and S2 (ESI†). The results show that all the peaks displayed in the patterns measured at room temperature closely match those in the simulated patterns from the structure data, indicating the phase purity of the solids. The PXRD pattern of desolvated solid **1a** shows a similar nature to that of the as-synthesized complex **1**. VT-PXRD patterns further indicate that the orders of the framework structure of complex **1** are retained upon complete removal of the guest water molecules. Furthermore, the PXRD patterns of **1** also remain intact upon immersion in water and boiling water, suggesting that no phase transition or framework collapse happens during these treatments. High stability of the coordination frameworks against water as well as the effects of temperature is essential for future practical applications in gas storage. Compared to the PXRD pattern of complex **2**, some main peaks in the PXRD pattern of the desolvated solid **2a** disappear and a few peaks also shift. This result indicates that the host framework of **2** is unstable and may collapse once the guest solvent molecules are removed.

Adsorption properties

Complexes **1** and **2** were subjected to gas adsorption studies. The as-synthesized compounds **1** and **2** were activated by immersing in methanol (MeOH) for 3 days and then outgassed overnight under vacuum to generate the dehydrated forms **1a** and **2a**. For **1a**, a gas adsorption curve is shown in Fig. 9. However, N_2 adsorption of **2a** appeared to fail since the framework collapsed to form a nonporous structure once the

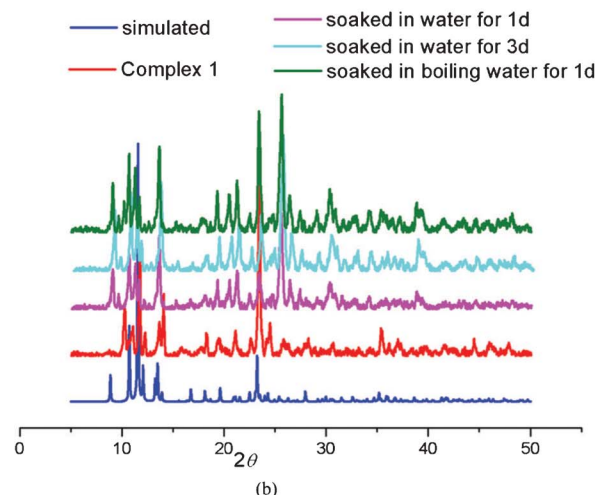


Fig. 8 (a) Variable-temperature PXRD patterns for **1**; (b) PXRD patterns of **1** obtained by different treatment in atmospheric water or boiling water for various durations.

guest solvent molecules were removed under vacuum.¹³ Thus, we merely discuss the details of adsorption properties for **1a** as follows.

The permanent porosity of **1a** was established by reversible gas sorption experiments. The N_2 adsorption isotherm at 77 K exhibits typical type I adsorption behavior, verifying its microporous nature. As shown in Fig. 9a, the maximum N_2 uptake is $100 \text{ cm}^3 \text{ g}^{-1}$ at 77 K, corresponding to a BET surface area of $370.2 \text{ m}^2 \text{ g}^{-1}$. The pore volume calculated from the maximum N_2 adsorption is $0.156 \text{ cm}^3 \text{ g}^{-1}$. By analyzing the crystal structure of **1a**, the internal diameter of the aperture is $4.1 \text{ \AA}$, which is too small to incorporate more N_2 molecules (the kinetic diameter for N_2 , $3.65 \text{ \AA}$). Moreover, there is only a one-dimensional channel system in the framework, and there are no additional channels along the crystallographic *a* and *b* axes in the case of the dehydrated framework of **1**. Therefore, at a low measuring temperature (77 K) the window of the channels is blocked by the entering N_2 molecules, which prevents other N_2 molecules from entering the pores.

H_2 adsorption isotherms were recorded under atmospheric pressures, which also showed type I behavior with good

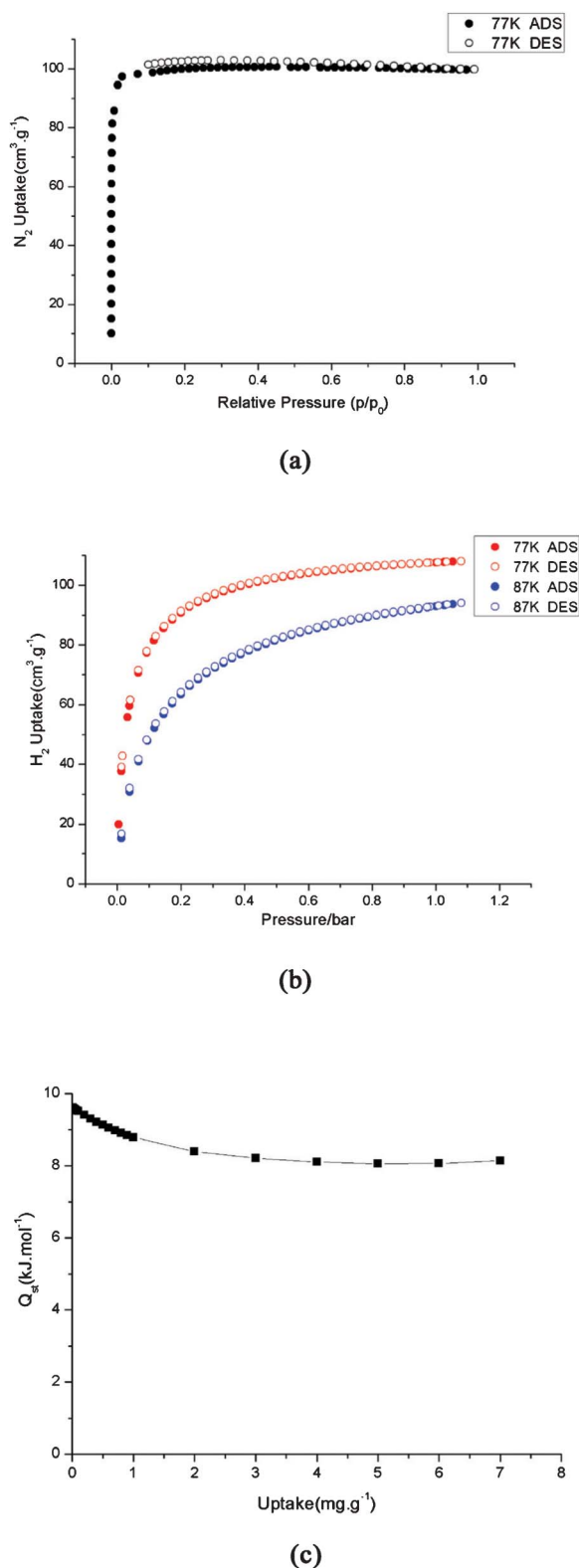


Fig. 9 (a) N₂ sorption isotherms of **1a** at 77 K; (b) H₂ adsorption isotherms of **1a** at 77 K and 87 K; (c) coverage dependency of the isosteric heat of adsorption for H₂ in **1a**.

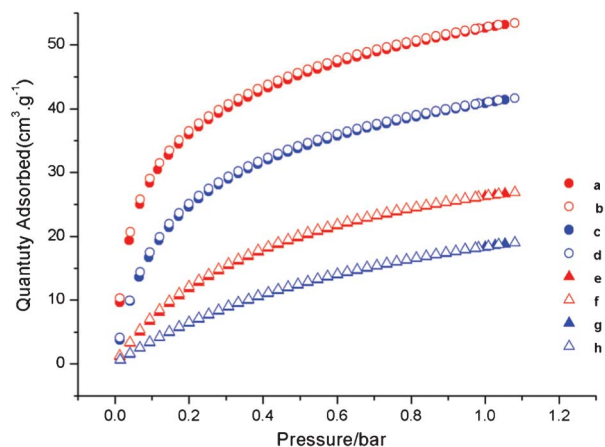


Fig. 10 The sorption isotherms of **1a**: CO₂ adsorption (a) and desorption (b) at 273 K; CO₂ adsorption (c) and desorption (d) at 300 K; CH₄ adsorption (e) and desorption (f) at 273 K; CH₄ adsorption (g) and desorption (h) at 300 K.

reversibility and no hysteresis. At 77 K and 1 bar, it had fully reversible uptake of 108 cm³ g⁻¹ (0.91 wt%), a value closing to those of recently reported MOFs at the same condition.¹⁴ To quantitatively compare their affinities to H₂ molecules, a second H₂ adsorption isotherm was measured at 87 K. The heat (Q_{st}) of H₂ adsorption is calculated by the Clausius–Clapeyron equation and the Q_{st} at zero coverage is estimated to be 9.6 kJ mol⁻¹ (Fig. 9c), higher than those of many famous porous materials, such as MOF-177 (4.4 kJ mol⁻¹), MOF-5 (5.2 kJ mol⁻¹), HKUST-1 (6.6 kJ mol⁻¹), or IRMOF-11 (9.1 kJ mol⁻¹).¹⁵ According to the nature of the pore structure, we consider that the high H₂ binding affinity observed in **1a** can be attributed to its relatively smaller pore size (the kinetic diameter for H₂, 2.89 Å).

To further explore its potential properties in gas sorption under ambient conditions, the adsorption isotherms of CO₂ and CH₄ for **1a** at 273 K and 300 K were measured (Fig. 10). The CO₂ uptake values at 1 bar are 53 cm³ g⁻¹ at 273 K and 42 cm³ g⁻¹ at 300 K. Coverage-dependent CO₂ adsorption enthalpies of **1a** were calculated according to the Clausius–Clapeyron equation. It shows that Q_{st} exhibits a maximum of 28.7 kJ mol⁻¹ at initial coverage (Fig. S3a in the ESI†).

The high Q_{st} for CO₂ benefited from the small channel size of **1**, because the framework–CO₂ interactions can be augmented in small pores. In addition, CH₄ adsorption was also measured at 273 and 300 K under 1 bar. The maximal CH₄ uptakes for **1a** are 27 cm³ g⁻¹ at 273 K and 19 cm³ g⁻¹ at 300 K. The Q_{st} of CH₄ adsorption at zero surface coverage is 20.5 kJ mol⁻¹ (Fig. S3b in the ESI†).

Photoluminescence properties

The photoluminescence of d¹⁰ metal complexes has been attracting intensive research interest, owing to their potential applications in chemical sensors, photochemistry and electroluminescent (EL) displays.¹⁶ Crystal photographs of complex **1** show that it exhibits strong blue emission under UV irradiation (Fig. S4 in the ESI†). The result can be further confirmed by the luminescent spectra. The solid-state emis-

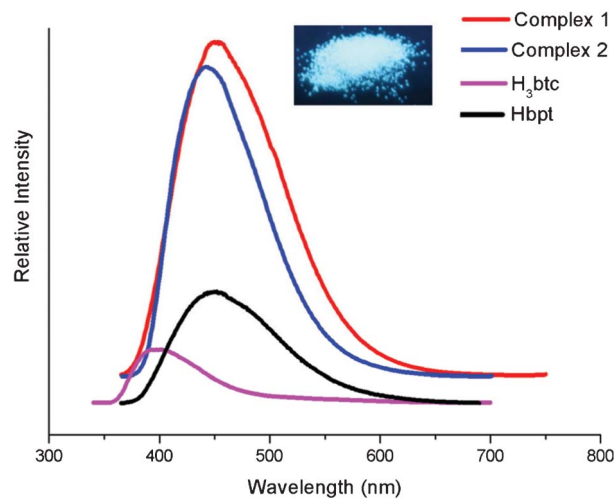


Fig. 11 Solid-state emission spectra of the free ligands and complexes **1** and **2** at room temperature. Inset: crystal photograph of **1** under UV irradiation.

sion spectra of complexes **1–2** with the ligand Hbpt and the O-donor auxiliary ligand H₃btc have been investigated at room temperature. As shown in Fig. 11, complexes **1** and **2** exhibit strong blue emission bands at 453 nm and 443 nm upon excitation at 350 nm, respectively, whereas the Hbpt and H₃btc ligands display very weak photoluminescence with emission maxima at 452 nm and 400 nm upon excitation at 350 nm. The emissions of **1** and **2** can probably be attributed to the interligand (π – π^*) charge transition.^{17,18} In addition, the fluorescent intensity for the coordination polymers is much stronger compared to that of the free ligands, possibly resulting from the enhancement of rigidity of the ligand after the coordination effect and the reduction of the loss of energy by radiation-less decay or vibrational motion.¹⁹ These observations indicate that complexes **1** and **2** may be considered as promising candidates for the exploration of fluorescent materials.

Conclusion

Two novel 3D microporous MOFs based on Hbpt and H₃btc ligands have been synthesized by using solvothermal method. Both of them are non-interpenetrated 3D frameworks with 1D channels. Topologically, complex **1** is a binodal (3,6)-connected three-dimensional network with rutile topology, while complex **2** is a (3,6,8)-connected three-nodal net with the Schläfli symbol of (4.6²)₂(4¹⁴.6¹².8²)(4⁴.6¹⁰.8). It is worth noting that complex **1** with small channels has unusual high stability and exhibits a remarkable heat of H₂ adsorption (9.6 kJ mol^{−1}), which may be expected for materials finding useful applications in hydrogen storage. In addition, the results of photoluminescent measurements indicate that **1** and **2** may be potential blue materials.

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